

Math 170A - Homework 2 - due October 18

Unless specified otherwise, show all your work.

Problem 1. If a day is sunny the probability that the next day will be rainy is $1/2$. If exactly k consecutive days have been rainy probability that the following day will be sunny is $\frac{1}{k+1}$. If today is sunny what is the probability that the next n days will all be rainy?

Problem 2. You are starting a new quarter at UCLA, and you enroll into classes in a way that for every offered class you decide to enroll it with probability p_1 , and not to enroll it with probability $1 - p_1$ (there are n classes offered). Decisions to enroll different classes are independent. Moreover, for every offered class you decide to purchase a textbook with probability p_2 , and not to purchase it with probability $1 - p_2$. Decisions to purchase different textbooks are independent. Also enrolling a certain class and buying the textbook for this class are also independent events. What is the probability of the event that you own the textbook for every class you enrolled?

Problem 3. Write out the expressions for (but no need to evaluate) the probabilities of the following events.

- (i) Head appears at most twice when a fair coin is tossed 8 times.
- (ii) A fair six sided die is rolled 5 times. Exactly 2 of the results are 1.
- (iii) A fair six sided die is rolled 10 times. At least 3 of the results are 1.

Problem 4. A clock tower at a certain university campus rings bell every hour. Every hour, it will ring once with probability $1/3$ and twice with probability $2/3$. The numbers of times the bell rings at different hours are independent. If you stay on campus for 4 hours (we hear bell on 4 occasions) what is the probability that we hear in total 7 rings.

Problem 5. You take care of n children and to keep them busy, you give to them (not to every child individually) n toys. You select these n toys from a big box which contains $2n$ different toys, n of which are blue and n of which are red.

The children don't like red toys, but you are colorblind and you can't tell which toy is red or blue, so you just select n toys at random. If children get any number of red toys (say, k red toys), they will return these k toys to you and complain. In this case you discard these k toys and you replace them with k toys chosen uniformly at random from the leftover toys in the box.

If they get red toys again they might complain but at this point you give up and they have to play with the toys they have.

- (i) What is the probability that children get all the blue toys in the first try?
- (ii) What is the probability that they get exactly 10 red toys in the first try?
What is the probability that you manage to replace this 10 with all blue toys?

- (iii) What is the chance they get all the blue toys eventually (either in the first try or in the second try)?

Problem 6. Ten people come to the party with their bikes, which are put in a separate room. At some point you realize you need more food, so you choose 4 guests and to each of them you give a bike from the separate room.

- (i) In how many ways can you do this?
(ii) What is the probability that every chosen person gets his/her own bike?

Problem 7. Assume that $0 \leq m \leq n$. Give a combinatorial proof that

$$\binom{n}{m} = \sum_{k=m}^n \binom{k-1}{m-1}.$$

(Hint: how many m -element subsets of $\{1, 2, \dots, n\}$ are there which have k as the largest element? Also, plug in $m = 2$ and simplify, what familiar formula do you get?)